Symbolic Dynamics

A Problem-Solving Exposition

Conventions

- An alphabet is a finite set $\mathcal{A}$.
- A *finite word* is a finite sequence of symbols from $\mathcal{A}$: $w = w_1 w_2 \dots w_n$, where $w_i \in \mathcal{A}$.
- The set of all finite words over $\mathcal{A}$, denoted by $\mathcal{A}^*$, together with the operation of concatenation, forms a monoid.
- An *infinite word* is an element of $\mathcal{A}^{\mathbb{N}}$, and a *bi-infinite word* is an element of $\mathcal{A}^{\mathbb{Z}}$.
- The set of all finite and infinite words over $\mathcal{A}$ is denoted by $\mathcal{A}^{\infty}$.

Definition 0.1 (Factor)**.** A factor of an infinite word w is a finite block of consecutive symbols in w.

Definition 0.2 (Shift)**.** The shift map $\sigma : \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ is defined by $(\sigma(w))_i = w_{i+1}$.

Exercise 0.1 (Bijectivity and Surjectivity of the Shift Map)**.** Show that the shift map σ is bijective on $\mathcal{A}^{\mathbb{Z}}$ and surjective on $\mathcal{A}^{\infty}$.

Exercise 0.2 (A Topology on $\mathcal{A}^{\mathbb{N}}$)**.** Let

$$e(x, y) = \min \{ n \geq 0 \mid x_i \neq y_i \},$$

with the convention that $\min \emptyset = \infty$, and define d as

$$d(x, y) = 2^{-e(x,y)}.$$

1. Show that d is a metric on $\mathcal{A}^{\mathbb{Z}}$.
2.

In the following, the topology we will consider is the one induced by the metric introduced in Exercise Exercise 0.2.

Exercise 0.3 (Continuity of the Shift Map)**.** Show that the shift map σ is continuous.

Example 0.1 (Even Shift)**.** Let X be the set of all infinite words over $\{0, 1\}$ where between every two 0s there are an even number of 1s.

Note that even shift is closed under the shift map.